

AYUSH ROY

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 GitHub |  LinkedIn |  Portfolio

EDUCATION

Sardar Vallabhbhai National Institute of Technology, Surat
Bachelor of Technology in Mathematics and Computing | CGPA: 7.83/10.00

Surat, India
2024 – 2028

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, Solidity, SQL, Bash, Rust

Systems & Runtime: Linux, Docker, Git, WebAssembly, EVM Internals, Multithreading, Distributed Systems

Program Analysis & Security: AST Parsing, CFG Construction, Static Analysis, Symbolic Execution, Fuzzing, Foundry, Hardhat

Data & Optimization: BERT, Sentence-Transformers (e5-base-v2), LLM-Assisted Tooling, XGBoost, SHAP, PyTorch, Differential Evolution, Particle Swarm Optimization

EXPERIENCE

FedEx Smart Center, IIT Madras

Chennai, India

Research Intern

May 2026 – July 2026

- Architected a dual-corpus data pipeline (218 documents spanning 13 institutional sources — APEDA, FSSAI, MPEDA, EU DG SANTE, US FDA, etc. — plus GDELT/YouTube/Reddit discourse) for a corpus-based NLP study on compliance barriers facing Indian agri-food MSME exporters, implementing a six-dimension data quality framework, OCR remediation, and a full decision/exceptions audit trail.
- Conducted independent, file-level pipeline audits that identified and eliminated fabricated data (80 synthetic news records and a hardcoded social-media fallback module) and a systemic entity-verification defect that had misclassified 95% of discourse records as registry-verified; redesigned the classification scheme into evidence-graded, registry-anchored verification tiers.
- Modeled compliance hurdles using BERTopic (e5-base-v2 embeddings, UMAP, HDBSCAN) across a four-locus × five-verification-logic analytical framework; diagnosed drift between the study’s research-question framing and its original protocol, and authored a 17-step, acceptance-tested corrective roadmap to restore methodological alignment.

Threat Prism

Remote

Cybersecurity Intern

Mar 2025 – Apr 2025

- Built Python automation scripts for network reconnaissance, port scanning, and OSINT-based security assessment workflows.
- Developed phishing-simulation tooling for internal security-awareness programs.

PROJECTS

EigenTrade: Quant Signal System | *Python, XGBoost, Optimization*

- Built a modular algorithmic trading pipeline from scratch using XGBoost, processing historical market data with rolling-window evaluation and SHAP-based interpretability analysis.
- Developed the underlying predictive logic using advanced evolutionary optimization techniques, specifically Differential Evolution (DE) and Particle Swarm Optimization (PSO).
- Quantified optimization success by increasing the predictive accuracy of buy/sell signals from 79% to 86% following DE and PSO implementation.

EigenVM — EVM Bytecode Interpreter | *Rust, WebAssembly*

- Built an Ethereum Virtual Machine bytecode interpreter from scratch in Rust, compiled to WebAssembly for browser-based execution.
- Implemented core EVM execution semantics — stack and memory model, opcode dispatch, and control flow — as a standalone runtime independent of existing EVM libraries.
- Originally developed as a systems-programming coding challenge; since extended with public documentation and a working demo.

Smart Contract Vulnerability Scanner | *Python, Solidity*

- Designed a modular security-analysis pipeline integrating AST parsing, CFG construction, and symbolic execution to detect 15+ vulnerability classes in Solidity contracts.
- Built a hybrid detection engine combining static rules with LLM-assisted execution-path validation to auto-generate gas-optimized patches, cutting manual audit time by 40%.
- Extended detection coverage for reentrancy and access-control flaws using fuzz-assisted validation across 10,000+ generated adversarial transactions per block.

Decentralized Remittance System | *Solidity, React, Web3*

- Architected a cross-border payment gateway on the Base L2 network using CBDC-based settlement, reducing transaction fees by 95% and achieving sub-10-second settlement finality.
- Implemented a trustless multi-signature escrow mechanism using OpenZeppelin contract primitives to secure funds during cross-border routing.
- Achieved 95%+ test coverage using Foundry and mainnet forking for end-to-end transaction-lifecycle simulation.

ACHIEVEMENTS & PROFILES

Competitive Programming: LeetCode Rating: 2043 (Knight Badge, Top 2.05%)

Profiles: LeetCode | HackerRank